

CS & IT ENGINEERING



Operating System

File System

Lecture -1

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Recap of Previous Lecture



Topic

Multilevel Paging

Topic

Frame Allocation



Topics to be Covered



Topic

Inverted Paging

Topic

File System



Topic : Question

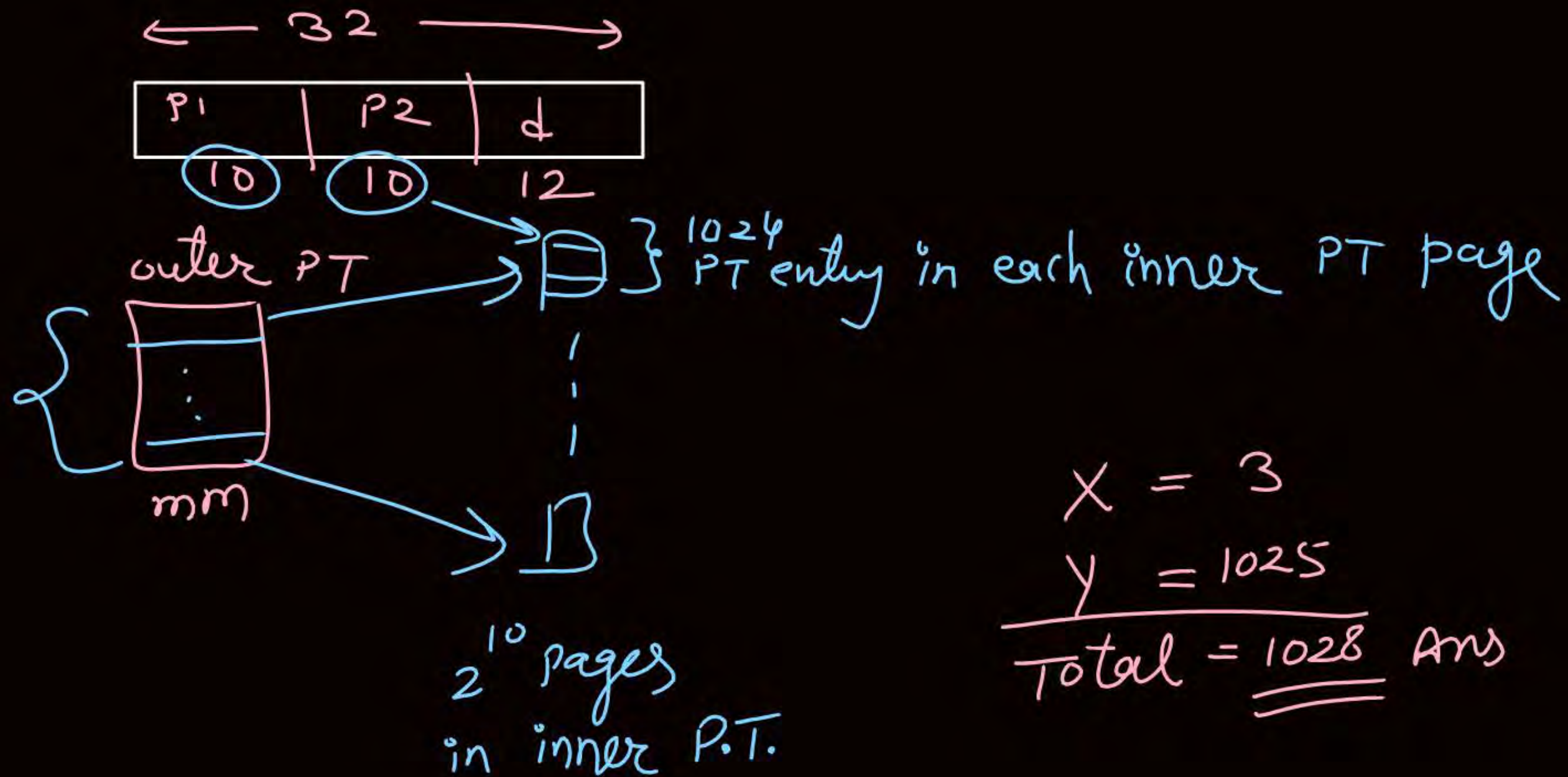
[GATE-2024]



#Q. Consider a 32-bit system with 4 KB page size and page table entries of size 4 bytes each. Assume $1 \text{ KB} = 2^{10}$ bytes. The OS uses a 2-level page table for memory management, with the page table containing an outer page directory and an inner page table. The OS allocates a page for the outer page directory upon process creation. The OS uses demand paging when allocating memory for the inner page table, i.e., a page of the inner page table is allocated only if it contains at least one valid page table entry.

An active process in this system accesses 2000 unique pages during its execution, and none of the pages are swapped out to disk. After it completes the page accesses, let X denote the minimum and Y denote the maximum number of pages across the two levels of the page table of the process.

The value of $X + Y$ is 1028?



$$\begin{array}{r}
 X = 3 \\
 Y = 1025 \\
 \hline
 \text{Total} = \underline{\underline{1028}} \text{ Ans}
 \end{array}$$



Topic : Inverted Page Table



in normal paging $\Rightarrow$ so many invalid entries possible
 $\Downarrow$

but still we provide space for
then in P.T.

$\Downarrow$
P.T. becomes very large

$\Downarrow$
To reduce it $\Rightarrow$ use inverted P.T.

$\Downarrow$
no. of entries = no. of frames

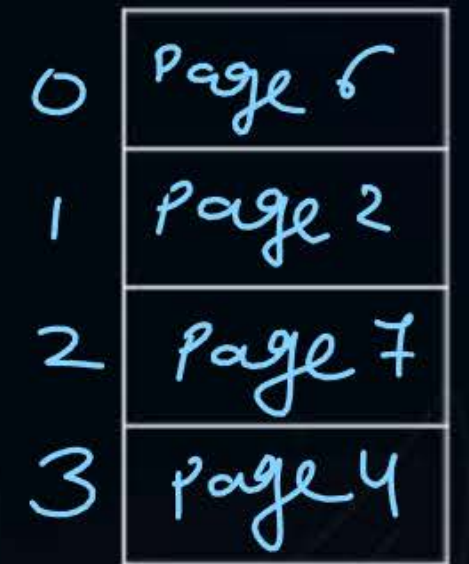
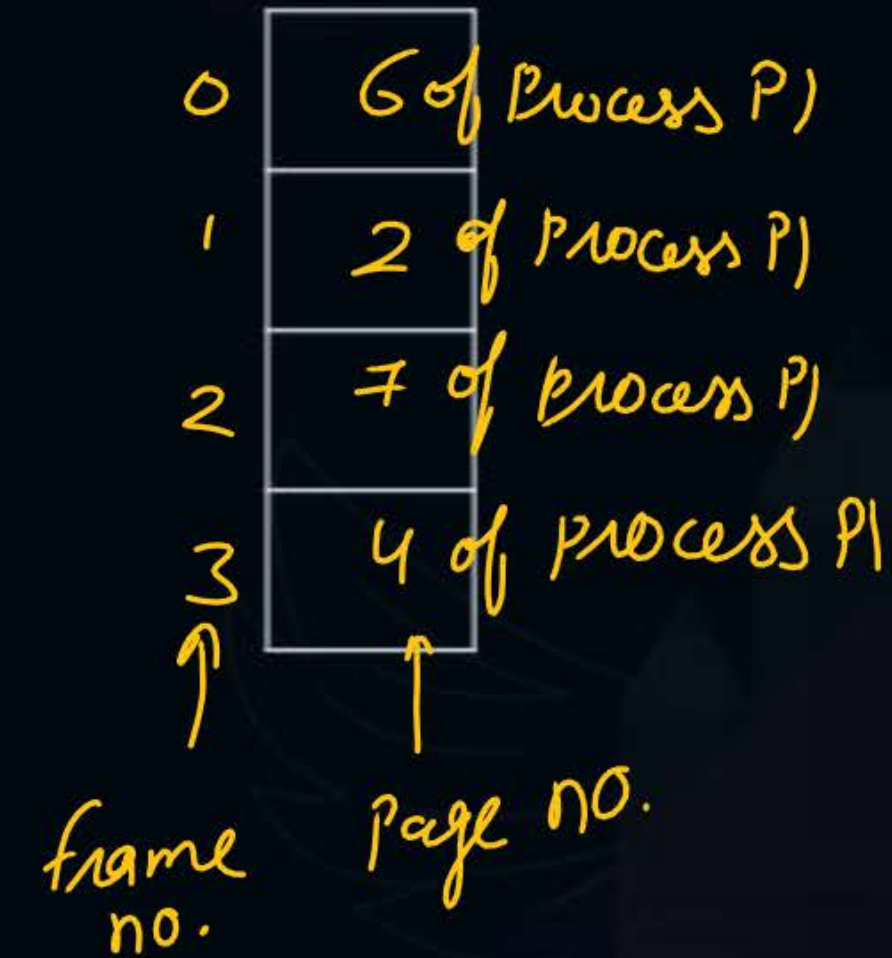
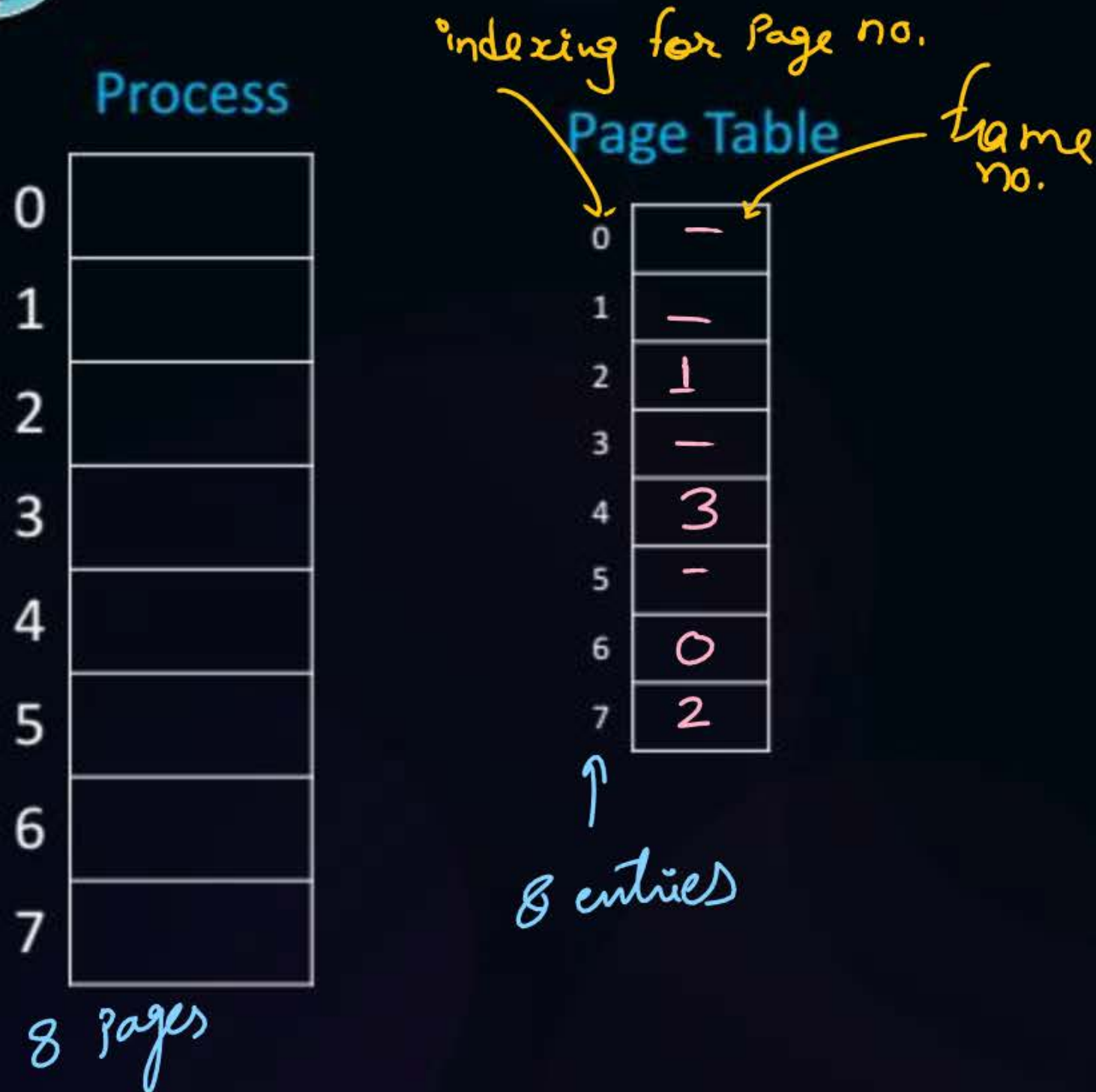


Topic : Inverted Page Table

single for all processes

Inverted Page Table

Physical Memory





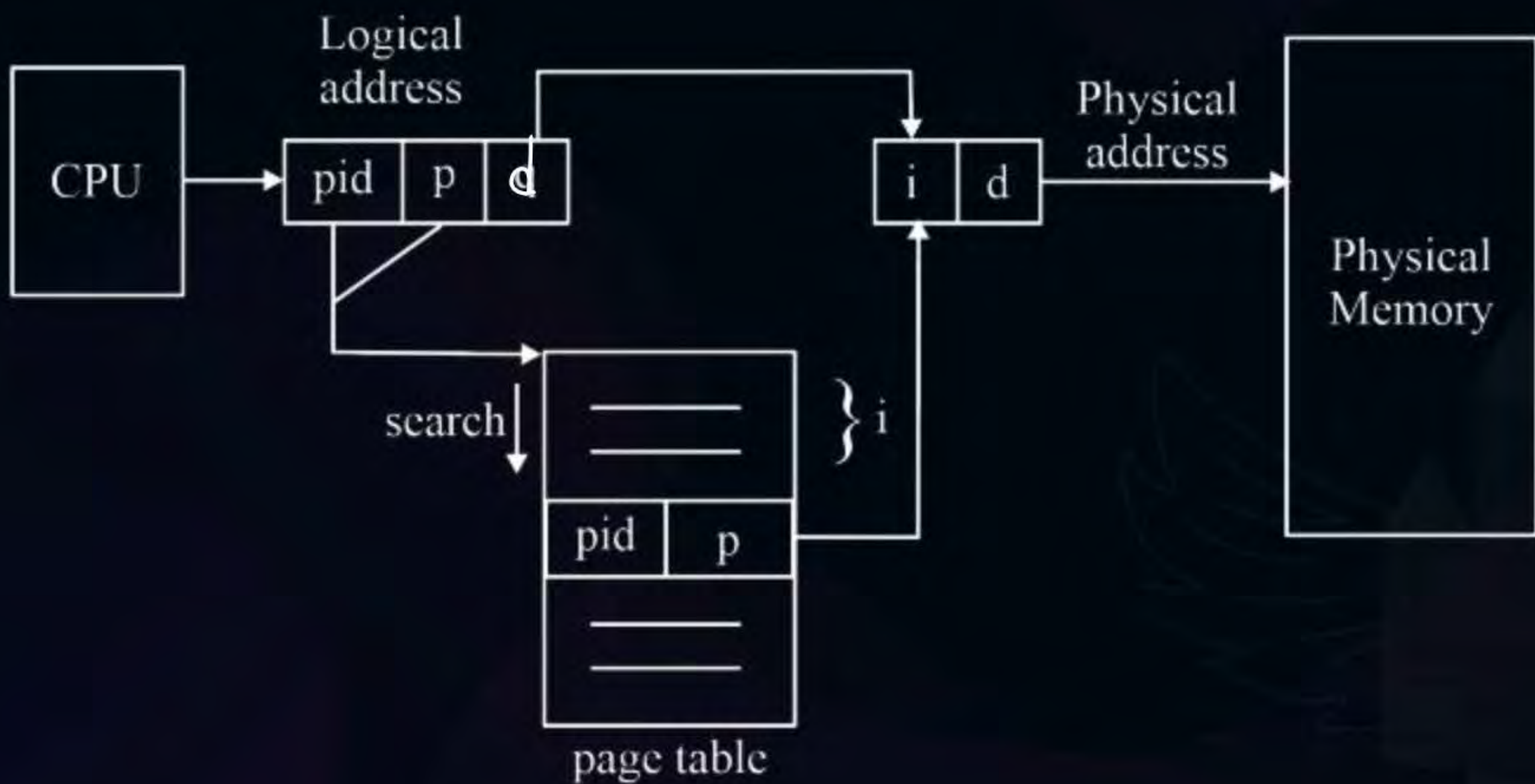
Topic : Inverted Page Table

Each entry in the page table contains the following fields:

1. Page number
2. Process id
3. Control bits
4. Chained pointer



Topic : Inverted Page Table





Topic : Inverted Page Table

Advantages & Disadvantages:

1. Reduced memory space
2. Longer lookup time
3. Difficult shared memory implementation



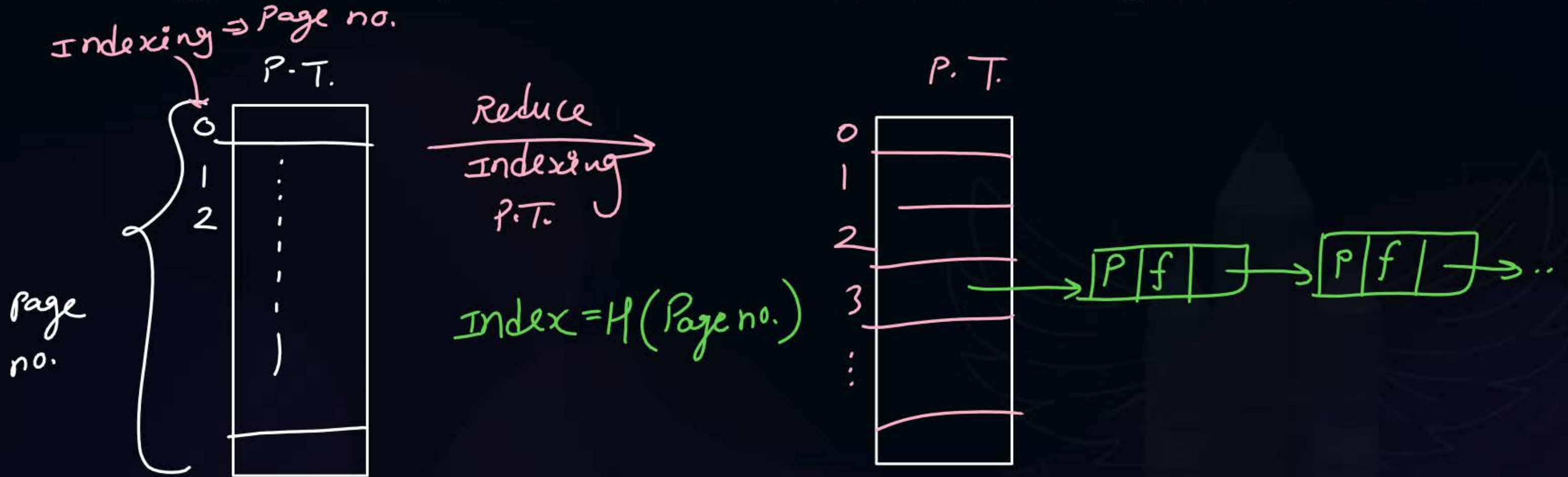
Topic : Hashed Page Table

→ One P.T. for one process

To reduce P.T. size

In this virtual page, the number is hashed into a page table

This Page table mainly contains a chain of elements hashing to the same elements.





Topic Hashed Page Table

Each element mainly consists of :

1. The virtual page number
2. The value of the mapped page frame.
3. A pointer to the next element in the linked list.

NEW
CHAPTER

file system



Topic : File

A file is a named collection of related information that is recorded on secondary storage.





Topic : File Attributes

1. Name
2. Extension
3. Size
4. Date
5. Author
6. Created, Modified, Accessed
7. Attributes: Read-only, hidden
8. Default Program
9. Security Details



Topic : File Directory

Collection of files

→ folder



Topic : File System

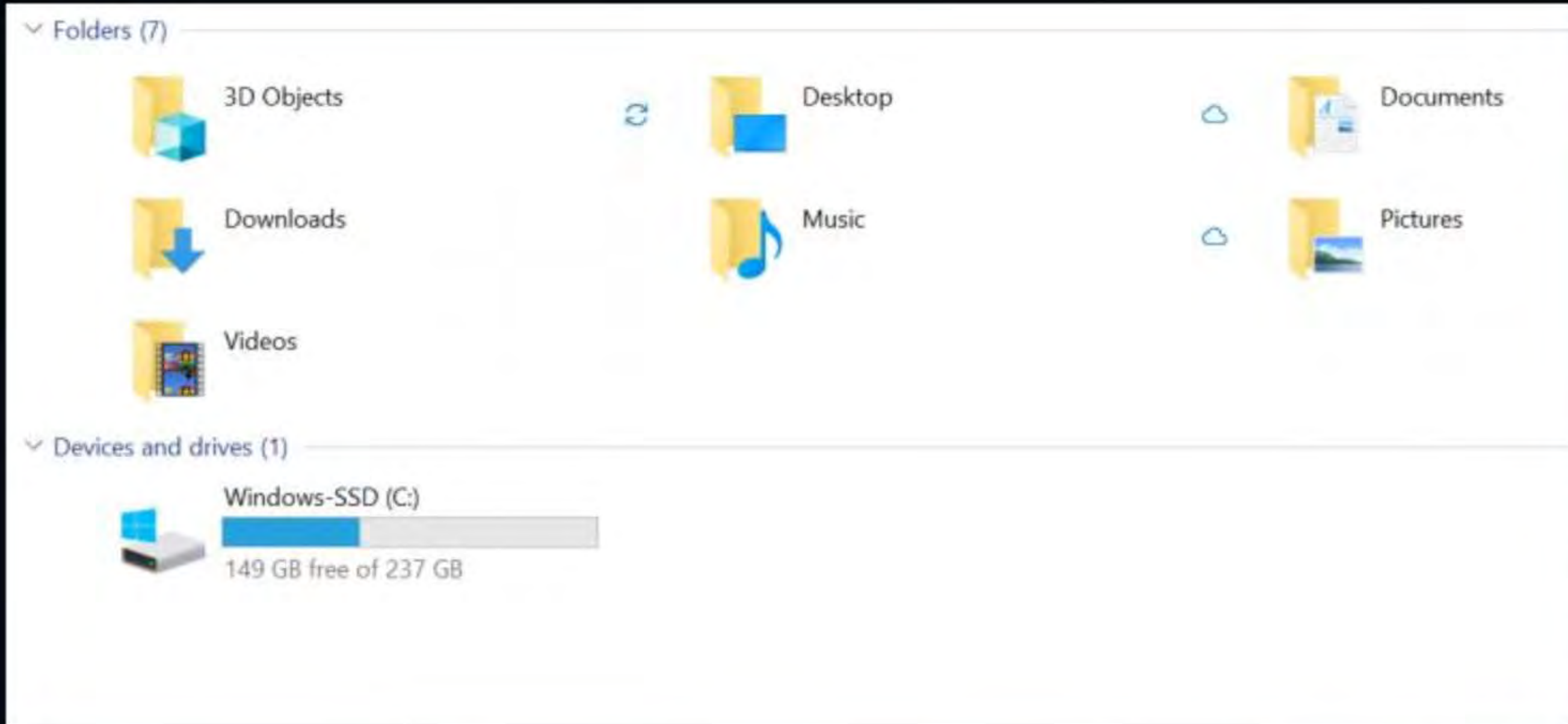


Module of OS which manages, controls and organizes files and related structures





Topic : File System





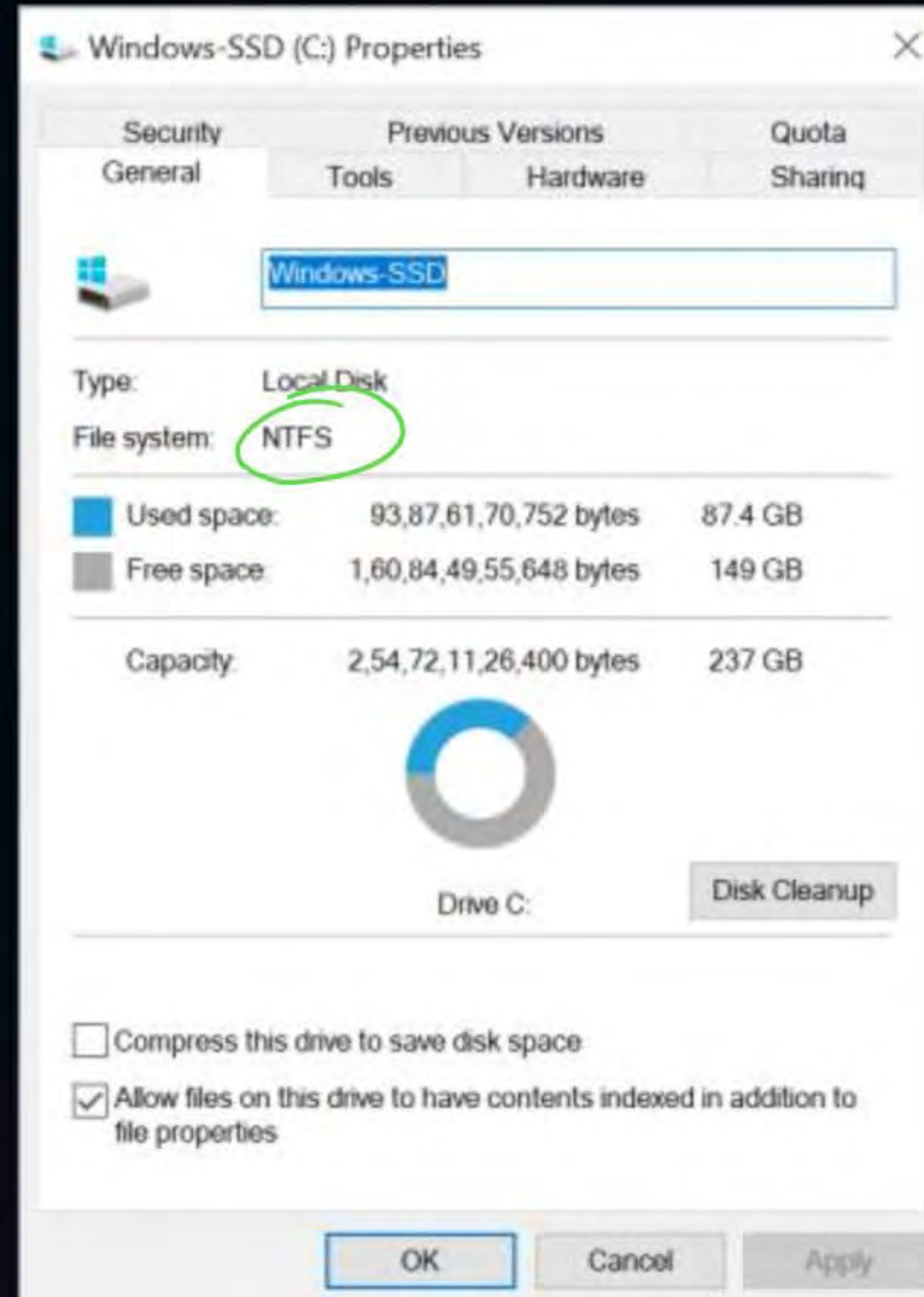
Topic : Types of File Systems



1. FAT32 ✓
2. NTFS ✓
3. HFS+ ✓
4. Ext2 / Ext3 / Ext4
5. Swap →



Topic : File System





Topic : Disk Formatting

Physical formatting
(low-level)

creating tracks & sectors

By manufacturer

Logical formatting
(high level formatting)

↓
creating logical partitions
in the disk.

↓
drives C:\ E:\ ...

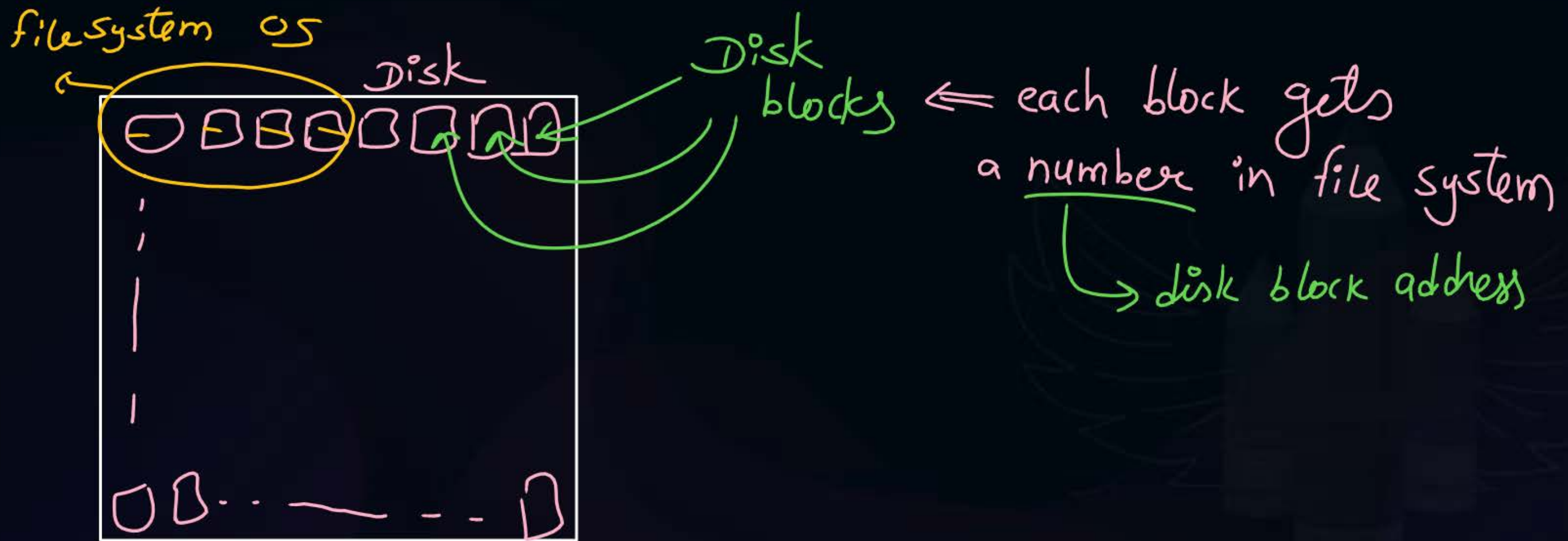
⇒ install OS in one drive

→ creating blocks



Topic : Disk Blocks

one disk sector known as disk block from file system's perspective.





Topic : Disk Blocks



disk block number = 16 bits

Number of disk blocks = 2^{16}

Size of each block = 1KB

Total Size of disk? = $2^{16} * 1KB = 2^{26} B$
= 64MB





Topic : Disk Blocks

Disk block address = 24-bits $\Rightarrow$ no. of blocks in disk = 2^{24}

Size of each block = 2KB

$$\begin{aligned}\text{Total Size of disk?} &= 2^{24} * 2KB \\ &= 2^{35} B \\ &= 32GB\end{aligned}$$





Topic : Disk Blocks



Total disk size = 256GB

Block Size = 2KB

Disk block address? $\Rightarrow$ 27 bits

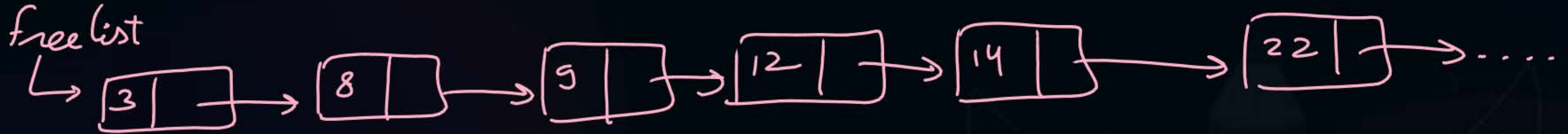
$$\begin{aligned}\text{no. of blocks in disk} &= \frac{256 \text{ GB}}{2 \text{ KB}} \\ &= 128 \text{ M} \\ &= 2^{27} \\ &\Downarrow \\ \text{address} &= 27 \text{ bits}\end{aligned}$$



Topic : Free Space Management

→ How file system maintains infoⁿ of all free blocks.

1. Free List $\Rightarrow$ A linked list in which address of free blocks are stored.
2. Bitmap Method $\Rightarrow$ one bit per block stored to denote a block is occupied or free. ex:- bit $\begin{cases} 0 & \text{free} \\ 1 & \text{occupied} \end{cases}$



block no. $\Rightarrow$ 3, 8, 9, ... are free blocks

Assume block no. $\Rightarrow$ starts from zero

Bitmap $\Rightarrow$ 1110111100110101111110

for block no.

block no. 3 is free



Topic : Free Space Management

1. No searching in free list, but in bitmap we search for first zero to find first free block
2. Free list is faster in allocating a free block
3. Free list size is variable, where as bitmap size is constant



Ques) A disk contains 16 blocks.
Bitmap { 1 free block
 0 occupied

→ 0011001010100001

$$\% \text{ of free blocks} = \frac{37.5}{100} \%$$

$$\text{free block} = 6$$

$$\% \text{ of free blocks} = \frac{6}{16} * 100\% = 37.5\%$$



Topic : Question

A particular disk unit uses a bit string to record the occupancy or vacancy of its disk blocks with '0' denoting vacant block and '1' denoting occupied block. A 32-bit part of this string has Hexadecimal value of A3D4B055. The percentage of occupied blocks on the disk for this part is ?

$$A3D4B055 \Rightarrow 1010001110101001011000001010101$$

$$\text{no. of occupied blocks} = 15$$

$$\begin{aligned} \text{\% of occupied block} &= \frac{15}{32} * 100\% \\ &= 46.875\% \end{aligned}$$

Ques) Disk has = 2^{25} blocks

Block size = 2KB

Total no. of blocks needed to store bitmap information
about free blocks 2048 ?

Solⁿ

$$\begin{aligned}\text{Bitmap size} &= 2^{25} \text{ bits} \\ &= \frac{2^{25}}{8} \text{ Bytes} \\ &= 2^{22} \text{ Bytes}\end{aligned}$$

$$\begin{aligned}\text{no. of blocks needed to} &= \frac{2^{22} \text{ B}}{2 \text{ KB}} \\ \text{store bitmap} &= 2^{11} \\ &= 2048\end{aligned}$$



2 mins Summary

Topic

Inverted Paging

Topic

File System

{
Track,
sector,
cylinder
}



Happy Learning

THANK - YOU

